

Related Works

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Annotated Bibliography: Games

There are a wide range of successful games and studies dedicated to educating students and professionals regarding the topics of computer networking and optimization. One augmented reality game was created and studied by Pereira and Viana [5] in which the researchers crafted an educational experience for students working in a space station where issues arise and puzzles must be solved. Pereira and Viana commented that “To solve the puzzle, our game requires the players to collect clues and utilize our knowledge of transport protocols, network addressing, and application layer protocols.” In addition, the researchers went a step further to host both an online and in-person iteration of the game with younger students, comparing the results. I found this game to be particularly useful because the developers used AFRAME, an AR/VR framework used consistently with P5JS, which is also the same software I’ll be using in my own project.

PeriHack is a tabletop-based card game designed to teach players about common network-based attacks, and defenses against them. The game is played out on a physical board representing specific network architecture with components such as a firewall, parking lot, company servers, etc. According to Dillon and Arushi [1], “In designing PeriHack our aim was two-fold for the red team player(s) we wanted to provide a way to illustrate different phases of the attack.” And “for the blue team, the objective was to make them aware of the countless possible vulnerabilities can be present in any given infrastructure.” I’m drawing inspiration from PeriHack’s use of a physical gameboard with components; and although I plan to incorporate threats into my product, I won’t be including a multiplayer component, and players won’t be playing as attackers.

Another interesting example of a useful network and cybersecurity-based tabletop tool is a tic-tac-toe game created by Kasim et al. [6], a game designed to demonstrate that AI based attackers may be nearly impossible to overcome by humans. On the surface, the users will see a tic-tac toe board; however, on the back end there is a complex attack and defend scenario where two docker containers are running competing programs; Kali Linux for the defender and Metasploit for the attacker. In addition, middleware is employed to translate the status of the attacks into simple Xs and Os. In terms of results, the researchers noted that “a highly trained AI Is nearly unbeatable by a human player and the only chance of a draw is if the human player executed the right move early gameplay.” This means that a human defender would have inside or prior knowledge of an impending attack. I thought this game was fascinating because of how simple, yet complex it appears to be so I’m interested in a similar approach to breaking down complex topics into simpler components.

Annotated Bibliography: Academic References

Nugraha et al. [3] published a collection of useful best practices and frameworks for incorporating computer networking topics into games. The authors note that “Conventional network learning generally studies theoretically and then performs simulations. However, when in real life, solving problems on a network also requires the ability to solve programs and search for the source of the problem that caused the network not to work properly.” This framework presents a series of considerations specific to game design regarding networking concepts such as user targeting, gameplay types, objectives and level stages. This source was useful to me when determining my target audience and goals for how I want the experience to be used.

A practical application of a framework for using games as a network planning tool can be found in Mabrouk and Naja’s [2] work with vehicular ad-hoc networks. In a healthcare scenario, “Chances of saving lives can increase remarkably if the ambulances are connected to emergency rooms (hospitals) so the vital signs of patients inside ambulances can be transferred to the hospitals in real time.” In their work, the researchers did not create a game but instead crafted a mathematical framework for how games could be used to model complex network architectures. In addition, Mabrouk and Naja account for a way to incorporate intrusion detection systems (IDS) in their framework which is highly relevant to my own project since I’ll be including an IDS in my product.

Ng and Hasan [4] conducted a systematic review of various cybersecurity learning tools and games, identifying strengths and flaws in each. For example, a common design weakness identified by the researchers was that “Firstly, cybersecurity training tools are often designed and prepared for only certain groups of users, such as professionals in the IT and cybersecurity field, workers in organizations, and students from various computing fields.” This research helped me compare the other existing games I had found during my research while incorporating aspects of my own project goals.

References

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